

Jyothir Anurag Vasu Chinnaboina

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Education

University of Wisconsin–Madison – *B.S. in Computer Science and Data Science*

Madison, WI

Awards: Dean's List (4 semesters)

Expected May 2026

Coursework: AI/ML, Reinforcement Learning, Algorithms, Operating Systems

GPA: 3.7/4.0

Statistics, Database Management, Data Structures

Experience

Software Developer Intern (Part-time) – *Vijay Rekha Life Sciences*

Aug 2025 – Present

- Deployed a neonatal screening lab service (FastAPI, PostgreSQL, React/TypeScript) processing 300+ samples daily with automated validation and abnormality flagging for 4 technicians.
- Automated 3+ hours of manual work into a single-click Python pipeline (pandas, xlsxwriter) that transforms raw mass spectrometry data for 40+ compounds into formatted Excel reports.
- Accelerated clinical report validation by 85% with a batch pipeline (pdfplumber, regex) that cross-validates patient data from 100+ PDF reports in a single run.

Mainframe Intern (Part-time) – *Wisconsin Department of Administration*

Jun 2025 – Present

- Reduced manual triage time by 90% with a Java 8 usage tracker (Python, z/OS) that parsed job logs into CSV and prioritized remediation follow-ups across 5+ teams.
- Eliminated 4+ hours/week of manual reporting via a Python orchestrator with YAML config, JSON Schema validation, Jinja2 templating, and automated email distribution.
- Deployed an IaC automation platform (Ansible, Terraform) adopted department-wide, streamlining 15+ batch job workflows across state mainframe systems.

Computer Science Tutor – *Undergraduate Learning Center, UW–Madison*

Sep 2024 – Aug 2025

- Tutored 600+ students across 8 CS courses and evaluated 160+ tutor candidates across 3 hiring cycles, designing technical interview questions for the Learning Center.

ERP Integration Intern – *United States Pharmacopeia*

Jun – Aug 2024

- Improved data retrieval time by 70% with an analytics dashboard (Python, pandas) using optimized SQL queries joining 5 tables to surface order hold trends.
- Automated quarterly stakeholder reports – previously 1 week of manual work – with an ETL pipeline (Matplotlib, Seaborn) rendering visualizations into Word, PPTX, and PDF.

Projects

Office Hours Queue | *FastAPI, ChromaDB, OpenAI API, MCP, Docker, React*

Sep 2025 – Present

Co-Founder · Advisor: Prof. Louis Oliphant (CS)

- Engineered an agentic code-analysis pipeline (MCP protocol, 9 custom tools) where an OpenCode LLM agent runs autonomously in sandboxed Docker containers, returning structured JSON reports.
- Built a production RAG assistant (FastAPI, ChromaDB, GPT-4o mini, Qwen3-Embedding-8B) with semantic search across 4 sources, cutting answer lookup to ~2s for 500+ students.
- Shipped LLM integration with retry logic, token budget management, and per-call cost tracking across dual providers (Qwen3 + OpenAI), logging every call for usage evals.
- Architected a 6-service containerized backend (React, Flask, FastAPI, ChromaDB, Redis, Traefik) with SSE streaming and GitHub Actions CI/CD for one-command deploys.

E-Beam 3D Printer Control System | *Python, Node.js, Express, GCP*

Jan – Aug 2025

Morgridge Institute for Research (Prof. Kevin Eliceiri)

- Architected a multi-threaded Python control system for an E-Beam 3D metal printer, integrating 6+ hardware subsystems over Serial and Modbus RTU with exponential-backoff retries.
- Engineered a remote monitoring dashboard (Node.js, Express, GCP) for researchers to track multi-day experiments, with log2-based downsampling to visualize weeks of sensor data.

AI/ML Algorithm Implementations | *Python, PyTorch, scikit-learn*

Sep – Dec 2024

- Implemented 9 ML algorithms from scratch including a LeNet-5 CNN (100-class scene classification), Q-Learning and SARSA RL agents (30K episodes), PCA, A* search, and Naive Bayes text classification.

Skills

Languages: Python, Java, JavaScript, TypeScript, Go, SQL, R, C, Bash, HTML/CSS

AI/ML: PyTorch, scikit-learn, ChromaDB, OpenAI API (GPT-4o), RAG, MCP (Model Context Protocol), LLM Agents, Embeddings, Semantic Search, pandas, NumPy

Backend & Infra: FastAPI, Flask, Django (familiar), Gin, Node.js/Express, React, Docker, Docker Compose, Redis, Traefik, GitHub Actions, Terraform

Databases & Cloud: PostgreSQL, Redis, Oracle DB, MySQL, SQLite, Google Cloud Platform

Other: REST APIs, SSE, WebSockets, OAuth2, JWT, Pytest, Agile/Scrum, Linux